Pan/Tilt Mini — printed parts & assembly

The 28BYJ-48 build, part by part. Chapter 1 identifies every piece on the 3D model from four angles; chapter 2 is the print list; chapters 3–5 cover the bought parts, settings, and build order.

At a glance

	Value
Envelope (assembled)	~100 x 94 x 126 mm
Printed parts	7 unique (9 prints incl. the x2 gears)
Bought parts	2 steppers, 4 bearings
Cut parts	2 steel shafts
Fasteners	M4 x2 (pan motor), M4 x2 (tilt motor), M3 grub x3, M3 x8 assorted
Est. print time (all parts)	~10–13 h total on one FDM machine
Est. filament	~120–150 g PLA

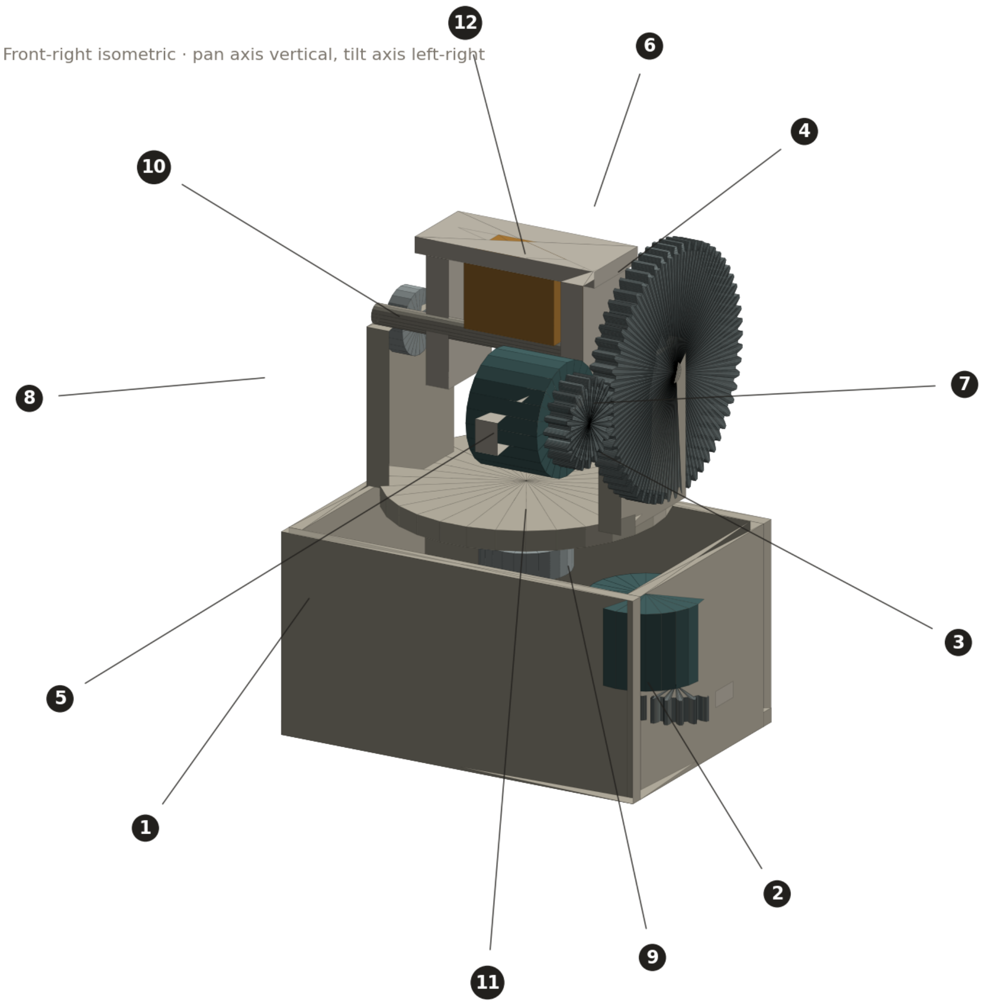
Print-time and filament figures are rough estimates from part volume, not slicer output — slice for your own machine before trusting them.

> These 3D views are built from the design geometry, not a photo of a finished unit. They are dimensionally consistent with the redesign study but are identification schematics: use them to know which part is which, and the print table plus your slicer for the real dimensions.

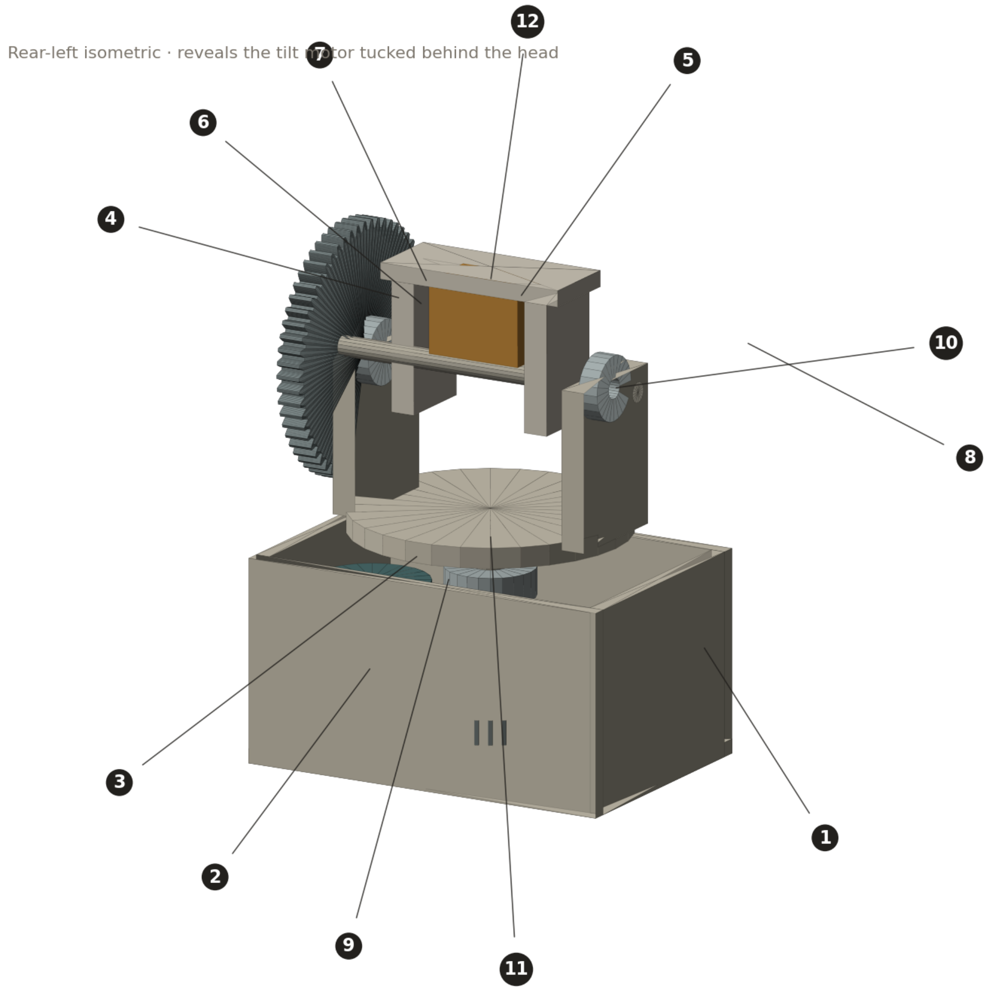
1. The assembly, identified

Every part carries the same number on all four views and in the print list. Printed parts are marked [print], bought parts [buy], cut stock [cut], and your payload [user].

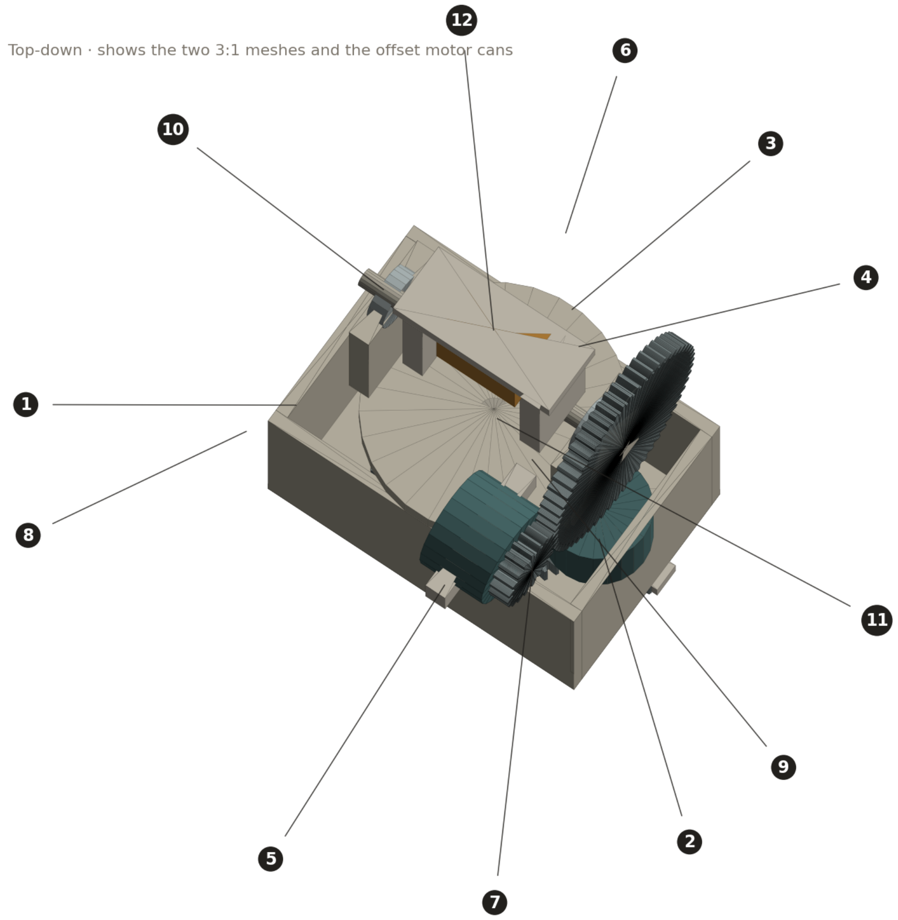
- **1** Base shell [print]
- **2** Pan-motor bracket [print]
- **3** Pan platform + tilt walls [print]
- **4** Yoke [print]
- **5** Tilt-motor bracket [print]
- **6** Spur gear 60T (x2) [print]
- **7** Spur gear 20T (x2) [print]
- **8** 28BYJ-48 stepper (x2) [buy]
- **9** 608ZZ bearing (x2) [buy]
- **10** 625ZZ bearing (x2) [buy]
- **11** Steel shaft rod (x2) [cut]
- **12** Payload (your sensor / pointer) [user]



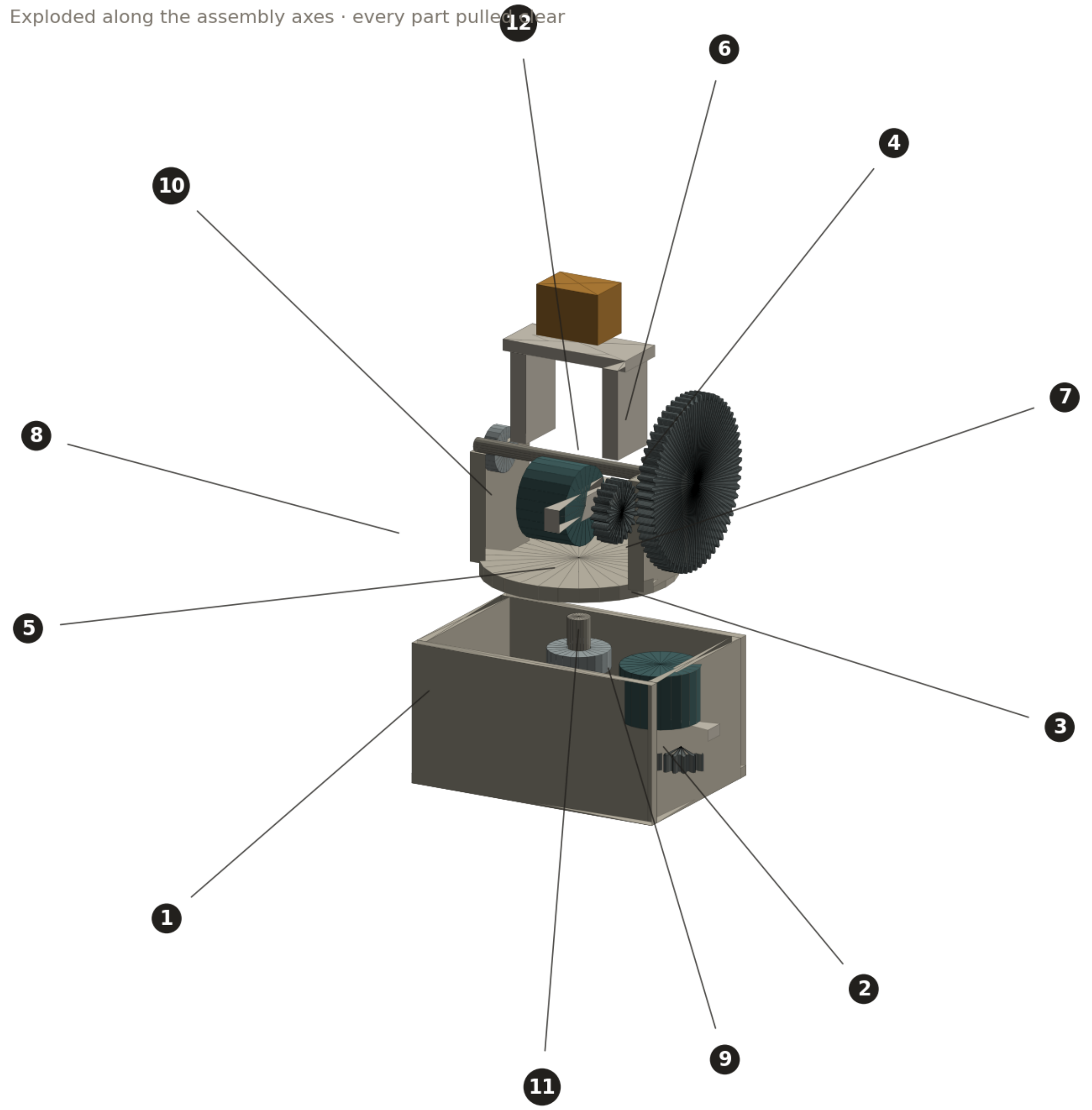
View A — Front-right isometric. Pan axis vertical through the base; tilt axis runs left–right through the head. The pan gear train (6, 7) and one stepper (8) live inside the base (1); the tilt gear (6) sits outboard on the right.



View B — Rear-left isometric. This angle exposes the tilt stepper (8) and its bracket (5) tucked 40 mm behind the tilt axis, and the back of the pan platform (3).



View C — Top-down. Shows both 3:1 meshes and how each stepper can (8) is clocked so its offset output shaft sits on the gear centre distance.



View D — Exploded. The definitive identification view: every part pulled clear along its assembly axis. Stack order from the base up is 1 → gear/bearings/motor → 3 → tilt shaft + 6/7/8 → 4 → 12.

What each numbered part is

#	Part	Type	What it is and where it sits
1	Base shell	print	The fixed foundation. An open-top tray, ~94 mm across, that houses the pan gear, both 608 bearings, and the pan stepper. Its top face carries the pan bearing bores and the pan-motor bracket.
2	Pan-motor bracket	print	A small two-tab plate that fixes the pan 28BYJ-48 inside the base with its shaft pointing down into the pan gear. Bolts to the base floor; the motor bolts to it with 2x M4.
3	Pan platform + tilt walls	print	The rotating structure — everything above the pan bearing turns with it. One printed piece: a disk keyed to the top of the pan shaft, plus two upright walls that carry the tilt shaft and both 625 bearings. This is the largest single print.
4	Yoke	print	Carries the payload and pivots on the tilt axis. Keyed to the tilt shaft with a D-flat and an M3 grub screw. Merges the original file's 'Cam Holder' and its separate pivot shaft into one member.
5	Tilt-motor bracket	print	Hangs the tilt 28BYJ-48 from one tilt wall, 40 mm behind the tilt axis so the head stays short. The motor bolts to it with 2x M4.
6	Spur gear, 60T (x2)	print	The driven gear of each 3:1 stage: module 1.0, 20° pressure angle, 62 mm outside diameter, 6 mm face. Two identical prints — one bored dia 8 with a D-flat for the pan shaft, one bored dia 5 for the tilt shaft.
7	Spur gear, 20T (x2)	print	The pinion of each stage, pressed on a stepper shaft. Same module and face as the 60T; 22 mm OD; dia 5 double-D bore with 3.0 mm flats to match the 28BYJ-48 shaft. Two identical prints.
8	28BYJ-48 stepper (x2)	buy	The two drive motors, geared 1/64 internally. One pans, one tilts. Bought, not printed — measure yours, as clones vary. Drives via a ULN2003 board (not shown).
9	608ZZ bearing (x2)	buy	8 x 22 x 7 skate bearings, spaced 16 mm apart in the base, carrying the pan shaft. The spacing gives the head its resistance to tipping.
10	625ZZ bearing (x2)	buy	5 x 16 x 5 bearings in the two tilt walls, carrying the tilt shaft.
11	Steel shaft rod (x2)	cut	Ground steel rod cut to length: dia 8 for the pan axis, dia 5 for the tilt axis. A flat is filed on each for its grub screws. Do not print these.
12	Payload	user	Your sensor, camera module, or laser pointer — up to ~15 g, mounted in the top of the yoke. Not part of the mechanism; shown to fix the balance point.

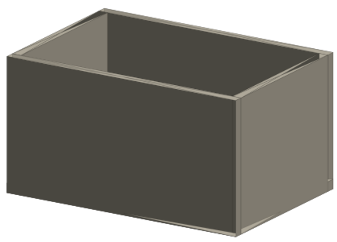
2. Parts to 3D-print

Seven unique parts, nine prints. Everything below prints in PLA on a 0.4 mm nozzle with no exotic settings. The orientation column is the one that matters most — especially the gears.

#	Part	Qty	Bounding box (mm)	Print orientation	~Time
1	Base shell	1	94 x 64 x 50	open side up, as it sits	3.0 h
2	Pan-motor bracket	1	42 x 16 x 12	flat	0.3 h
3	Platform + walls	1	68 x 68 x 42	platform down on the bed	2.8 h
4	Yoke	1	48 x 20 x 36	cross-member down, arms up	1.4 h
5	Tilt-motor bracket	1	40 x 8 x 14	flat	0.4 h
6	Spur gear 60T	2	62 x 62 x 6	FLAT on the bed, axis vertical	1.1 h ea
7	Spur gear 20T	2	22 x 22 x 6	FLAT on the bed, axis vertical	0.3 h ea

The one rule that decides whether the gears work: print them flat, axis vertical, so layer lines run across the tooth face. A gear printed on its side has layer-line seams along the flanks and will mesh roughly and wear fast.

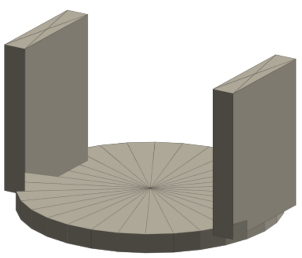
Individual parts



1 Base shell

94 x 64 x 50 mm

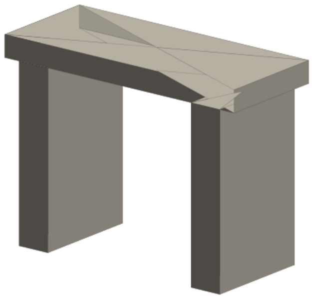
Open-top tray. Bores: 2x dia 22 (608 seats) on the top face, on the pan axis; 2x M4 for the pan-motor bracket. Calibrate the 22 mm bores with a coupon first.



3 Platform + tilt walls

68 x 68 x 42 mm

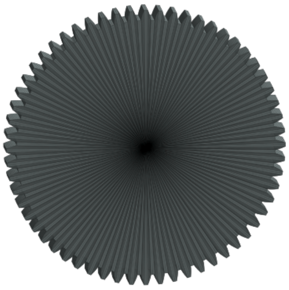
One piece. Central dia-8 D-bore keys to the pan shaft; the two walls carry dia-16 (625) bores on the tilt axis. Largest single print — print platform-down, walls up.



4 Yoke

48 x 20 x 36 mm

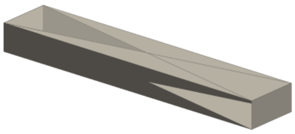
Straddles the tilt shaft. Dia-5 D-bore + M3 grub on one arm keys it to the tilt rod. Payload pocket in the top cross-member. Print cross-member down to avoid arm supports.



6 Spur gear 60T (x2)

62 x 62 x 6 mm

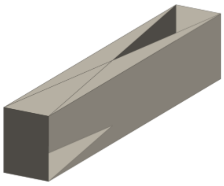
m1.0, 20T/60T stage, driven gear. Bore dia 8 + D-flat (pan copy) or dia 5 (tilt copy). Print flat, 4 perimeters. This face-on view shows the tooth form.



2 Pan-motor bracket

42 x 16 x 12 mm

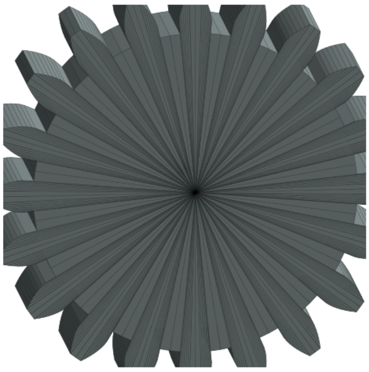
Two-tab motor plate, 35 mm hole centres for the 28BYJ-48. Prints flat in minutes.



5 Tilt-motor bracket

40 x 8 x 14 mm

L-plate that hangs the tilt motor behind the head. 35 mm hole centres. Prints flat.



7 Spur gear 20T (x2)

22 x 22 x 6 mm

The pinion. Dia-5 double-D bore, 3.0 mm across flats, to grip the stepper shaft without a grub screw. Print flat, 4 perimeters.

3. Bought and cut parts

Nothing here is printed. Buy or cut these; they define their own fits.

#	Part	Spec	Qty	Notes
8	28BYJ-48 stepper	5 V (or 12 V), dia 28 can, 5 mm offset D-shaft	2	Ships with a ULN2003 driver board. Measure the real shaft offset and can height before printing brackets 2 and 5.
9	608ZZ bearing	8 x 22 x 7	2	Pan axis. The most common bearing made.
10	625ZZ bearing	5 x 16 x 5	2	Tilt axis.
11	Steel rod	dia 8 x ~48 (pan), dia 5 x ~80 (tilt)	2	Ground/silver steel. File a flat where each grub screw lands.

Fasteners

- **2x M4** — pan motor to bracket 2.
- **2x M4** — tilt motor to bracket 5.
- **3x M3 grub** — 60T gear to pan shaft, yoke to tilt shaft, one spare on the pinion if you skip the D-fit.
- **~8x M3** — bracket-to-base and bracket-to-wall fixings; exact count depends on how you tab the brackets.

4. Print settings

Setting	Value	Why
Material	PLA	Stiff, dimensionally stable, loads here are tiny.
Nozzle / layer	0.4 mm / 0.15 mm	0.15 mm keeps tooth flanks clean.
Perimeters (gears)	4	At m1.0 that is essentially a solid tooth.
Perimeters (structure)	3	Plenty; only the yoke sees any bending.
Infill	25–30%	Structure. Gears end up near-solid from perimeters alone.
Supports	none if oriented as listed	The orientations above avoid overhangs.
Gear orientation	flat, axis vertical	Layer lines across the flank, not along it.

> **Calibrate the bearing bores before the big prints.** FDM prints holes undersize by a machine-specific amount. Print a coupon with dia 21.8 / 22.0 / 22.2 mm holes, find the one that presses a 608 in firmly, and use that number in the base. Repeat with dia 15.8 / 16.0 / 16.2 for the 625 in the walls.

5. Assembly order

Build from the pan axis outward. Test each stage's rotation by hand before adding the next.

- **Base (1).** Press the two 608s (9) into the top bores. Bolt bracket (2) inside.
- **Pan shaft (11, dia 8).** Press the 60T gear (6) on near the bottom; slide the shaft up through both 608s. It should spin freely.
- **Pan motor (8).** Press a 20T pinion (7) on its shaft, bolt it to bracket (2), and check the pinion meshes the pan gear with a trace of backlash.
- **Platform + walls (3).** Key it to the top of the pan shaft (grub screw). The whole upper structure now pans.
- **Tilt bearings + shaft.** Press the two 625s (10) into the walls. Press the 60T gear (6, dia-5 bore) onto the tilt rod (11, dia 5) at the outboard end, slide the rod through both walls.
- **Tilt motor (8).** Pinion (7) on its shaft, motor onto bracket (5), bracket to the wall. Check the tilt mesh.
- **Yoke (4).** Key it to the tilt shaft between the walls (grub screw). It should tilt smoothly through its full arc.
- **Payload (12).** Fit last, so you can balance it about the tilt axis — that keeps holding torque near zero.

> **Wiring across the pan joint.** The tilt motor rides on the platform, so its 4-wire cable has to cross the pan axis. Leave a service loop and cap pan travel, or fit a small slip ring at the pan axis if you need continuous rotation — the base has room for one.